Week 10: Consciousness: The Hard Problem

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Even a complete account of what a system does may leave open whether there is something it is like to be that system. This week introduces the concept of phenomenal consciousness, the arguments that it resists physical explanation and the main positions in the debate [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to distinguish phenomenal from access consciousness and to state the hard problem and the explanatory gap. They should be able to reconstruct the zombie and knowledge arguments and test their validity. Finally, they should be able to characterise physicalist, dualist and illusionist responses.

Lecture

What it is like

Nagel argued that an organism has conscious mental states if and only if there is something it is like to be that organism [1]. A bat perceives through echolocation, and however much is learned about its neurophysiology, it seems impossible to know what it is like for the bat. The subjective character of experience appears to escape the objective descriptions of science. Block distinguished two concepts that are often run together [4]. Access consciousness concerns information that is available for reasoning, report and the control of action. Phenomenal consciousness concerns experience itself, what it is like. Much of the confusion about machine consciousness comes from moving between the two.

The hard problem and the explanatory gap

Chalmers separated the easy problems of consciousness from the hard problem [2]. The easy problems concern functions: discrimination, integration of information, report, attention and the control of behaviour. They are easy only in the sense that it is clear what an explanation would look like, namely a mechanism that performs the function. The hard problem is why the performance of these functions is accompanied by experience at all. Levine had described the

same difficulty as an explanatory gap [5]. Even if pain is correlated with a neural process, nothing in the process makes it intelligible why it should feel the way it does.

Chalmers argued that the hard problem shows consciousness to be a fundamental feature not reducible to physical processes [6]. His main argument concerns zombies: beings physically identical to people but without experience. If zombies are conceivable, they are possible, and if they are possible, physicalism is false. Jackson's knowledge argument makes a parallel case [3]. Mary is a scientist who knows all physical facts about colour vision but has lived in a black-and-white room. When she first sees red, she learns something new, so there are facts that are not physical.

Responses

Physicalists answer in several ways. Some deny that zombies are genuinely conceivable once the physical facts are fully understood. Others accept conceivability but deny that it implies possibility. On their view, phenomenal concepts pick out physical states in a special way, so Mary gains a new concept or ability rather than knowledge of a new fact. Dennett offered a more radical response [7]. On his view, the idea of an inner theatre in which experiences are presented to a self is an illusion. Once all functional facts are explained, nothing further remains to be explained. This line of thought, now often called illusionism, treats the hard problem as a problem about why people believe in phenomenal properties. The interactive lab asks students to state their own position and tests it for consistency with the arguments.

Why the debate matters for AI

If consciousness is a matter of function, a machine that performs the right functions is conscious. If it is not, no amount of functional similarity settles the question. The inverted spectrum, shown in Figure 10.1, makes the point vivid. Two agents could name every colour in the same way while their inner qualities differ, and no behavioural test would reveal the difference. Critics have argued that the structure of human colour space makes such an undetectable inversion impossible, but the thought experiment shows why behavioural evidence underdetermines claims about experience. Week 11 turns to scientific theories that try to bridge the gap.

> ***Pause and reflect.*** *Is the difference between access and phenomenal consciousness something you can detect in your own experience, or only a distinction between concepts?*

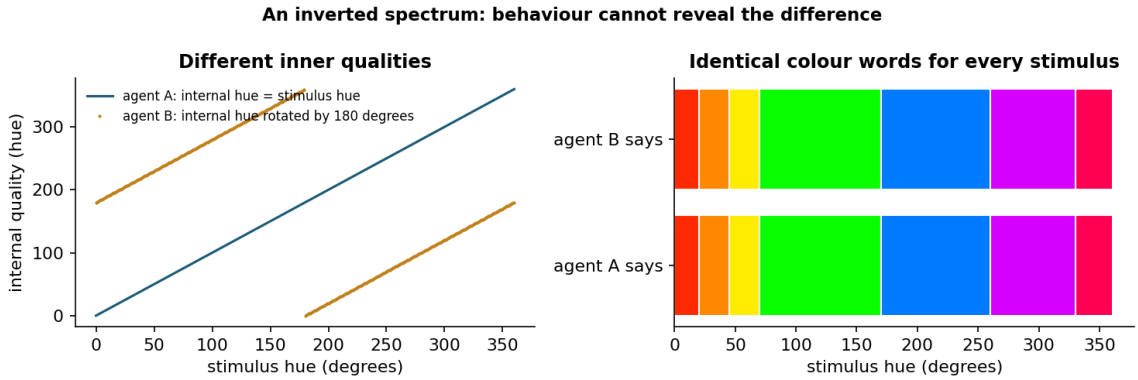


Figure 10.1. An inverted spectrum. Agent B's inner qualities are rotated relative to agent A's, yet both use the same colour words for every stimulus. Behaviour therefore cannot reveal the difference.

Hands-on components

Interactive lab: Part A and Part B check the zombie and knowledge arguments against your judgements. Part C compares your answers with the typical commitments of several positions. Part D shows an inverted spectrum in which two agents behave identically [3, 6]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-10/lab.html

Colab notebook: It checks the validity and consistency of the zombie and knowledge arguments by truth tables and simulates two agents with rotated inner colour codes whose behaviour is identical [3, 6]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-10/NB10_hard_problem.ipynb

Reflection prompts

1. Which premise of the zombie argument do you find weakest, and what would it take to convince you otherwise?
2. If illusionism is right, what exactly would a machine need in order to be conscious in the only sense that exists?
3. Does the inverted spectrum threaten the use of behavioural tests for machine consciousness more than for human consciousness? Why?

Weekly task and submission

What to do. Reconstruct one argument of this week and take a position on it. First, write the zombie argument or the knowledge argument as numbered premises and a conclusion, and check its validity with the truth tables of the Colab notebook. Second, identify the premise you reject, or accept with the most doubt, and give your reason. Third, explain what your position implies for the possibility of conscious AI.

Where to look. Lecture: sections "The hard problem and the explanatory gap", "Responses" and "Why the debate matters for AI". Interactive lab: Part A for the zombie argument, Part B for

the knowledge argument and Part C, which compares your answers with several positions. Colab notebook: Section 1 with Exercises 1 and 2.

Example. A physicalist could accept that Mary learns something when she first sees red, but deny that she learns a new fact. On this view, she gains a new concept of a physical state. Whether an AI system could be conscious would then depend on which physical or functional properties it has.

What to submit. About 500 words, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

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[6] Chalmers, D. J. (1996). The Conscious Mind: In Search of a Fundamental Theory. Oxford University Press.

[7] Dennett, D. C. (1991). Consciousness Explained. Little, Brown and Company.